IMPACT & BENEFITS

Impact & Benefits Report

Expected value, theory of change, evaluation metrics and responsible deployment

IMPACT-04

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VAYU NEXA AI | Alchemy Architects | SIH 2026

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1. Executive statement

The potential impact of AAKAASH ADHINAYAK is best framed as better-quality planning evidence: improved visibility into the age and origin of records, earlier detection of data defects, faster comparison of constrained alternatives, transparent rejection of invalid schedules and more reproducible evaluation. These benefits are hypotheses until measured against a clearly defined baseline. The current public site inspection does not establish actual realised impact.

Evidence note: The public URL resolves to a page titled "AAKAASH ADHINAYAK - Verified Intelligence for Adaptive Resource Planning", but the available research interface did not expose navigable page text, runtime interactions, source code, API traces or test results. Therefore this report distinguishes observed deployment metadata and the team-supplied blueprint from proposed design and unverified implementation. It is not a source-code audit, security assessment, or certification of the live system.

2. Intended beneficiaries

Beneficiary	Expected benefit (hypothesis)	What must be measured later
Planner / analyst	Less manual collation, clearer constraints and quicker comparison of alternatives.	Paired task timing, error counts and explanation quality.
Reviewer / supervisor	Traceable inputs and transparent validation outcomes.	Sample audit: reconstruct a selected run from its evidence bundle.
Data steward	Faster discovery of missing, stale, duplicate and conflicting data.	Known-defect fixture recall/precision and time-to-detect.
Researcher / educator	Repeatable experiments and safe exploration of infeasible cases.	One-click replay, fixed seeds/versions and reproducible results.
Institution / pilot sponsor	A measured view of planning quality, auditability and integration costs.	Pilot metrics, documented limits and total cost assumptions.

3. Benefit logic / theory of change

Inputs	Activities	Outputs	Outcomes / impact
Source files, public data, domain constraints, documented test scenarios.	Validate, normalise, optimise, independently check, simulate and explain.	Issue lists, candidate plans, validator reports, scenario diffs and audit exports.	Lower defect escape, lower time to compare options, stronger reproducibility and better-informed human review.

The chain only holds when every step works. A dashboard that reports a data-quality count but cannot trace it to underlying records does not deliver the stated benefit. An optimiser without independent checks may produce an attractive but invalid plan. A language-model explanation without source grounding can reduce rather than increase trust.

4. Quantitative evaluation plan

Feature	Quantitative method	Success criteria	Reporting
Data-defect detection	Create labelled fixture with missing, malformed, duplicate, stale and conflicting records.	Report precision/recall by issue type; no invented result.	Report confusion counts and missed defect examples.
Hard-constraint integrity	Use a test set with known feasible and infeasible plans; mutate valid plans to create violations.	Zero invalid candidates marked VALIDATED in the acceptance suite.	Publish test count, failures and software version.
Plan quality vs baseline	Run a naive documented baseline and optimiser on identical inputs.	Compare objective, assignment count, delay, utilisation and runtime.	Do not call one approach better unless criteria and result support it.
Dynamic response	Inject the same disruption across repeated runs.	Measure replanning runtime and difference from original schedule.	Disclose machine, scenario size, warm/cold state and run counts.
Human task time	Time users solving the same task with current method and prototype.	Report median and spread across participants; no claim from one trial.	Use consent and avoid unnecessary personal data.
Prediction quality	Holdout evaluation on a documented public prognostics dataset.	Compare baseline and candidate with MAE/RMSE or appropriate classification metrics.	Record dataset/version/splits and avoid claims about military engines.
Evidence traceability	Sample run exports and try to reconstruct each result.	All sampled results resolve to input snapshot, solver config and validator.	Report failures and orphaned references.

5. Benefits by time horizon

Immediate - within the prototype

- Visible source state and freshness rather than ambiguous “live” labels.
- Faster diagnosis of bad input through structured issues and affected-record links.
- A candidate scheduling result that explicitly reports validated, review-required, infeasible or unknown.
- Scenario replay and evidence export that make demonstrations reproducible.

Medium term - during a pilot

- Measured reduction in manual collation or evaluation time if paired testing proves it.
- Better ability to compare trade-offs without concealing unassigned tasks or infeasibility.
- Improved review workflow through role-based review notes and audit trail.
- Documented understanding of data limitations and integration effort.

Longer term - only after institutional validation

- Possible integration with permitted enterprise data sources.
- Possible workflow gains in civilian maintenance, inventory or resource scheduling.
- Potential use as an educational/research platform for responsible, evidence-grounded decision support.

6. Accessibility, responsible AI and public value

The product should be usable by keyboard, have clear focus states, avoid relying on colour alone, respect reduced-motion preferences and provide text alternatives for charts/3D scenes. WCAG 2.2 offers an established accessibility baseline [R11]. The AI component should identify uncertainty, avoid fabricated records and remain subordinate to deterministic verification and human review. These choices improve comprehensibility and reduce the risk that polished visuals are mistaken for evidence.

The project can be described as thematically aligned with innovation and resilient infrastructure under UN Sustainable Development Goal 9 [R14]. That is not evidence that the product has produced measurable sustainable-development impact.

7. Impact risks and unintended consequences

Risk	Possible harm	Mitigation
Automation bias	Users trust the optimiser or LLM without checking input validity.	Separate solver and validator; show source quality, assumptions and review state.
Misleading “live” status	Old/cached/fictional data may be mistaken as current.	Explicit data classes, timestamps, freshness thresholds and degraded states.
Metric gaming	Team reports a percentage improvement from cherry-picked scenarios.	Pre-register test scenarios, report negative outcomes and disclose baseline.
Sensitive data leakage	Uploaded workforce or operational information is exposed.	Use generic IDs, minimise collection, apply roles, retention rules and access tests.
Visual accessibility exclusion	3D effects hide information from some users or on low-end devices.	Accessible 2D list, text summaries, reduced motion and responsive layout.

8. Impact conclusion

Impact position: State benefits as hypotheses today. Establish impact by pairing a fixed baseline with repeatable scenarios and reporting observed values, variance, test conditions and known limitations. The strongest early metric is integrity: no invalid candidate is labelled validated. Time-saving and performance benefits come only after measurement.

References: [R3], [R5], [R7], [R11], [R14]. See the reference register for source details.